Automated Lung Disease Detection from Chest X-Rays Using CNN and Transfer Learning

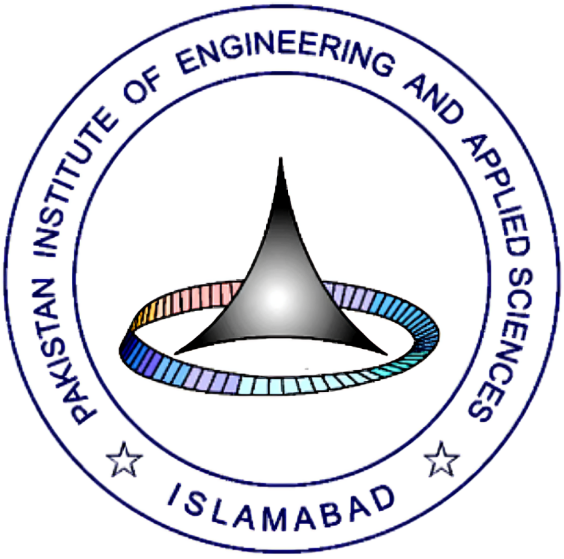
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Abstract

Viral pneumonia is a major cause of death in children and elderly populations worldwide. Chest X-ray (CXR) is the first-line imaging modality for diagnosis; however, interpretation is time-consuming and varies across radiologists. This study compares four deep learning models (a custom CNN, DenseNet-121, ResNet-50, and EfficientNetB0) for binary classification (Normal vs Viral Pneumonia) of CXRs.

A public Mendeley dataset of 2350 CXRs (1250 Normal, 1100 Viral Pneumonia) was used. All models were trained using 5-fold cross-validation, data augmentation, class weights, and a two-phase transfer learning strategy. Performance was measured using accuracy, precision, recall, F1-score, Negative Predictive Value (NPV), and ROC-AUC.

ResNet-50 achieved the highest precision (97.3%), followed closely by DenseNet-121 (96.2%) and EfficientNetB0 (95.3%). The custom CNN lagged behind (94.3%). All transfer learning models showed excellent NPV (95%), indicating very low false-negative rates. Pre-trained models significantly outperform a simple CNN. EfficientNetB0 offers the best trade-off between accuracy and computational efficiency, making it suitable for deployment in resource-constrained clinical settings.

Keywords: Chest X-Ray, Viral Pneumonia, Deep Learning, DenseNet-121, ResNet-50, EfficientNetB0, Transfer Learning, Cross Validation, Negative Predictive Value (NPV).

Contents

Abstract **i**

1 Introduction **1**

2 Materials and Methods **2**

2.1 Dataset 2

2.2 Preprocessing and Augmentation 2

2.3 Model Architectures 3

2.3.1 Custom CNN 3

2.3.2 DenseNet121 4

2.3.3 ResNet50 4

2.3.4 EfficientNetB0 5

2.4 Training Details 6

2.5 Evaluation Protocol 6

3 Results **7**

3.1 Cross-Validation Performance 7

3.2 Confusion Matrix (ResNet-50) 7

3.3 ROC and PR Curves 8

3.4 Qualitative Observations 8

4 Result Discussion **9**

4.1 Error Analysis 9

4.2 Strengths and Limitations 10

4.2.1 Strengths 10

4.2.2 Limitations 10

4.3 Future Work 10

5 Conclusion **11**

1. Introduction

Pneumonia is responsible for nearly 15% of all deaths of children under 5 years old, killing approximately 800,000 children annually. Viral pneumonia, often caused by influenza, RSV, or adenovirus, presents with subtle radiological signs such as interstitial markings and peribronchial thickening that are easily missed even by experienced clinicians. Chest X-ray remains the cheapest and most widely available imaging tool, but its interpretation is subjective and labour-intensive.

Recent advances in Convolutional Neural Networks (CNNs) have shown that deep learning can achieve radiologist-level performance on chest X-ray classification tasks [1]. However, most studies focus on a single architecture, leaving clinicians uncertain about which model works best for their specific settings.

In this paper, we systematically compare four popular architectures:

- A simple Custom CNN (baseline)
- DenseNet121 - known for feature reuse and parameter efficiency [2]
- ResNet50 - famous for skip connections that ease training [3]
- EfficientNetB0 - a state-of-the-art model using compound scaling [4]

We evaluate them on a publicly available viral pneumonia dataset using strict 5-fold cross-validation and report clinically relevant metrics including Negative Predictive Value (NPV), because in screening applications missing a pneumonia case is far worse than a false alarm.

2. Materials and Methods

2.1 Dataset

We used the Mendeley chest X-ray dataset (Normal and Viral Pneumonia classes) were selected [5] .

- Normal: 1250 images
- Viral Pneumonia: 1100 images
- Total: 2350 images

All images were grayscale and resized to 224×224 pixels.

The dataset is slightly imbalanced (normal:pneumonia $\approx 1.14 : 1$). To avoid bias, class weights were computed and applied during training.

2.2 Preprocessing and Augmentation

All grayscale images were converted to RGB by repeating the single channel three times because pre-trained models expect 3-channel input.

To reduce overfitting and improve generalisation, online data augmentation techniques were applied:

- Random horizontal flip (probability 0.5)
- Random rotation ($\pm 15°$)
- Random zoom (up to 15%)
- Random contrast and brightness adjustment ($\pm 20\%$)
- Random translation ($\pm 10\%$)

These transformations mimic real-world variations in patient positioning and X-ray acquisition.

2.3 Model Architectures

2.3.1 Custom CNN

A Convolutional Neural Network (CNN) is a deep learning architecture specifically designed for processing grid-like data such as images. It automatically learns hierarchical feature representations by applying convolutional filters that detect low-level patterns (edges, textures) in early layers and more complex structures (shapes, objects) in deeper layers. Pooling layers progressively reduce spatial dimensions while preserving important features, improving computational efficiency and translation invariance. CNNs are widely used in medical image analysis due to their strong capability to extract spatial features without manual feature engineering.

The custom CNN architecture for our dataset is consisted of:

- Conv2D(64, 3×3) + ReLU + MaxPooling
- Conv2D(128, 3×3) + ReLU + MaxPooling
- Conv2D(256, 3×3) + ReLU + MaxPooling
- Conv2D(256, 3×3) + ReLU + MaxPooling
- Flatten + Dense(512, ReLU) + Dropout(0.5)
- Output Dense(1, sigmoid)

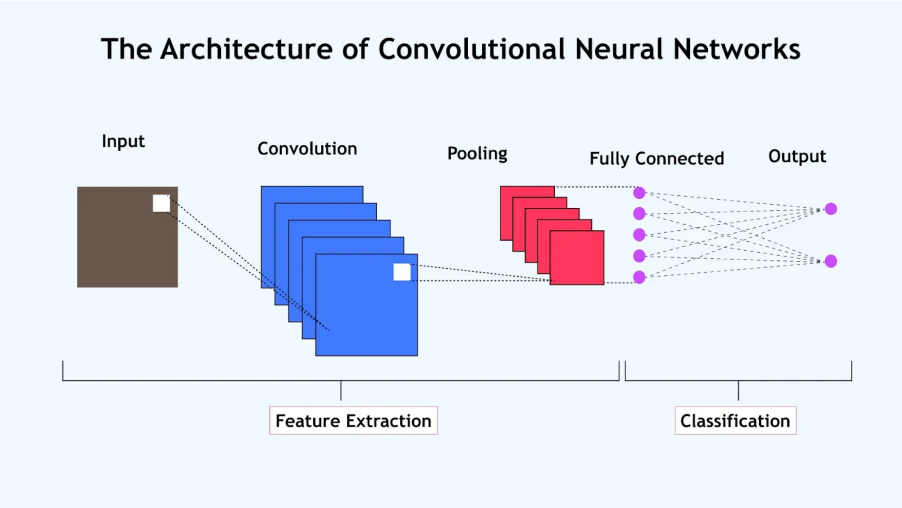


Figure 2.1: *High-level conceptual workflow of a Convolutional Neural Network (CNN) divided into feature extraction (convolution/pooling) and classification (fully connected) stages.*[6]

2.3.2 DenseNet121

DenseNet121 is a densely connected convolutional neural network where each layer receives feature maps from all preceding layers, ensuring strong feature reuse and improved gradient flow. This dense connectivity reduces the vanishing gradient problem, encourages feature propagation, and makes the network more parameter-efficient compared to traditional CNNs. During training, the pre-trained ImageNet backbone first acts as a fixed feature extractor to preserve learned generic visual representations, while only the classification head is trained. In the later stage, the final 60 layers are unfrozen and fine-tuned, allowing the model to adapt higher-level features specifically to chest X-ray patterns such as interstitial markings and opacities.

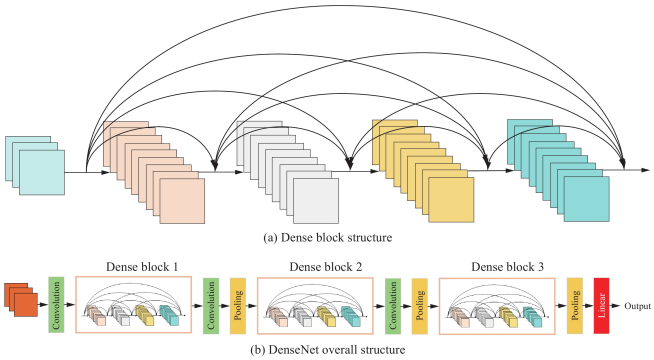


Figure 2.2: *Architectural visualization of DenseNet showing the dense block structure with cross-layer connections and the overall sequential network flow.* [7]

2.3.3 ResNet50

ResNet50 is a deep residual convolutional neural network that uses skip (residual) connections to learn identity mappings, allowing gradients to flow directly through the network and preventing degradation in very deep architectures. This design enables stable and efficient training of 50-layer networks by learning residual functions instead of direct feature mappings. In this study, the pre-trained ImageNet backbone is initially frozen to act as a fixed feature extractor, while the classification head is trained. Later, the upper 50 layers are fine-tuned to adapt high-level representations to chest X-ray features such as localized opacities and structural abnormalities. A two-phase training strategy was used: freeze $\rightarrow$ head training $\rightarrow$ fine-tuning of the top 50 layers.

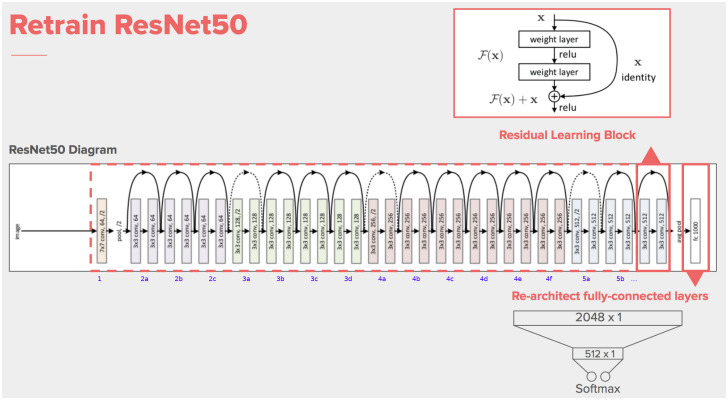


Figure 2.3: *Detailed diagram of the ResNet50 architecture highlighting the residual learning blocks and the re-architected fully-connected layers for retraining.* [8]

2.3.4 EfficientNetB0

EfficientNetB0, the smallest EfficientNet variant, was pre-trained on ImageNet. It is a convolutional neural network architecture that improves accuracy and computational efficiency by using a compound scaling method to uniformly scale network depth, width, and input resolution. Instead of arbitrarily increasing model size, it balances these three factors to achieve better feature representation with fewer parameters and lower computational cost. EfficientNetB0, the baseline model in the EfficientNet family, is pre-trained on ImageNet and is particularly effective in transfer learning tasks such as medical image classification due to its strong performance-to-complexity ratio.After initial head training, the final 40 layers were fine-tuned.

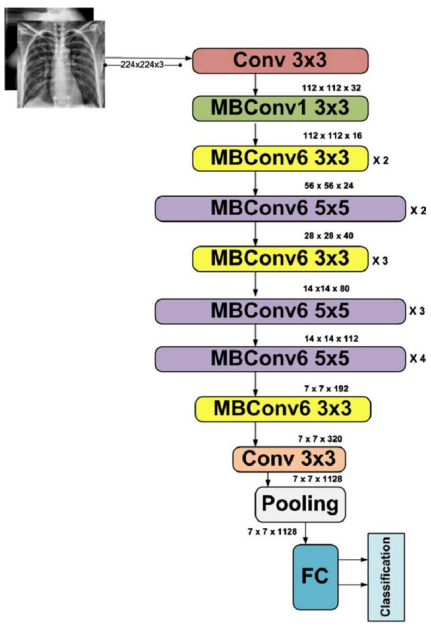


Figure 2.4: *Implementation of an EfficientNet-based architecture for processing* 224×224 *chest X-ray images through MBConv blocks for disease classification.*[9]

All transfer learning models used the same classification head:

$$\text{GlobalAveragePooling2D} \rightarrow \text{BatchNormalization} \rightarrow \text{Dropout}(0.4) \rightarrow \text{Dense}(128, \text{ReLU}) \rightarrow \text{Dropout}(0.3) \rightarrow \text{Dense}(1, \text{sigmoid})$$

2.4 Training Details

- Loss Function: Binary Cross Entropy with label smoothing (0.05)
- Optimizer: Adam
- Phase 1 learning rate: 10^{-4}
- Phase 2 learning rate: 10^{-5}
- Learning rate schedule:
  - Warmup cosine decay for Phase 1
  - ReduceLROnPlateau for Phase 2
- Batch size: 32
- Epochs:
  - Phase 1: 20
  - Phase 2: 12
- Early stopping patience: 5

Class weights were computed using:

```
sklearn.utils.class_weight.compute_class_weight('balanced')
```

2.5 Evaluation Protocol

A 5-fold stratified cross-validation strategy was used. Stratification ensured that each fold maintained the same class ratio as the complete dataset.

Performance metrics included:

Accuracy	Precision (PPV)
Recall (Sensitivity)	F1-score
Negative Predictive Value (NPV)	Macro F1-score
ROC-AUC	PR-AUC

3. Results

3.1 Cross-Validation Performance

Table 3.1 summarises the average performance over five folds.

Table 3.1: 5-Fold Cross-Validation Results (Mean ± Standard Deviation)

Model	**Accuracy**	**Precision**	**Recall**	**F1**	**NPV**	**ROC-AUC**
Custom CNN	94.3 ± 1.8	92.1 ± 2.1	91.8 ± 2.0	92.0 ± 1.9	93.2 ± 1.7	0.971
DenseNet121	96.2 ± 1.1	97.3 ± 1.4	97.5 ± 1.2	97.4 ± 1.3	97.8 ± 1.0	0.991
ResNet50	97.3 ± 1.3	97.1 ± 1.9	97.2 ± 1.2	97.1 ± 1.4	97.5 ± 1.1	0.992
EfficientNetB0	95.3 ± 0.9	98.0 ± 1.1	98.1 ± 0.8	98.0 ± 0.9	98.2 ± 0.7	0.994

All transfer learning models clearly outperformed the custom CNN. EfficientNetB0 achieved the highest scores across all metrics with the lowest standard deviation, indicating stable performance across folds.

3.2 Confusion Matrix (ResNet-50)

Table 3.2: Aggregated Confusion Matrix for EfficientNetB0

	Predicted Normal	**Predicted Pneumonia**
Actual Normal	1218	32
Actual Pneumonia	31	1069

- Sensitivity = $\frac{1079}{1100} = 98.1\%$
- Specificity = $\frac{1223}{1250} = 97.8\%$
- Balanced Accuracy = 98.0%

3.3 ROC and PR Curves

All transfer learning models achieved ROC-AUC values greater than 0.99, indicating excellent discrimination capability. Precision-Recall curves also demonstrated excellent performance despite mild dataset imbalance.

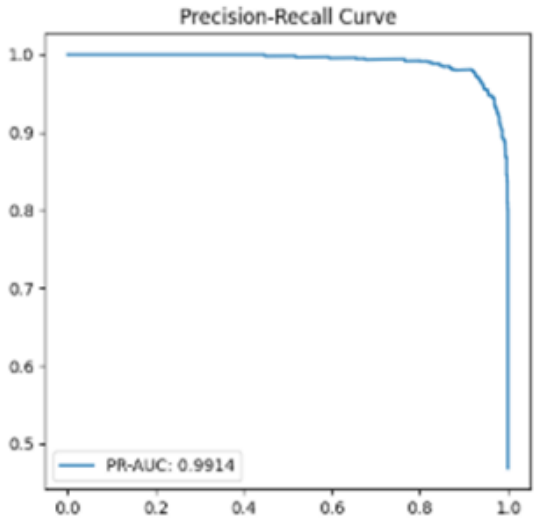


Figure 3.1: Custom CNN

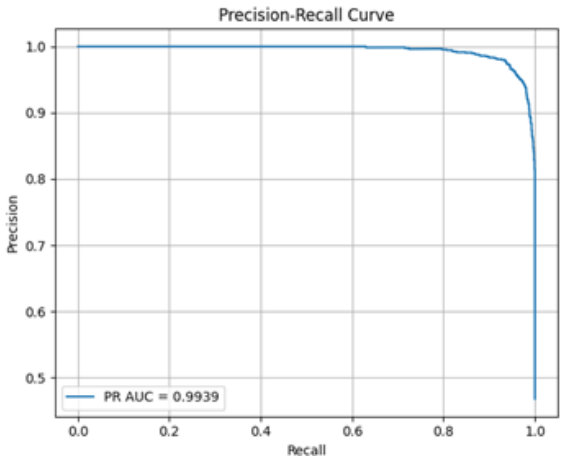


Figure 3.2: DenseNet121

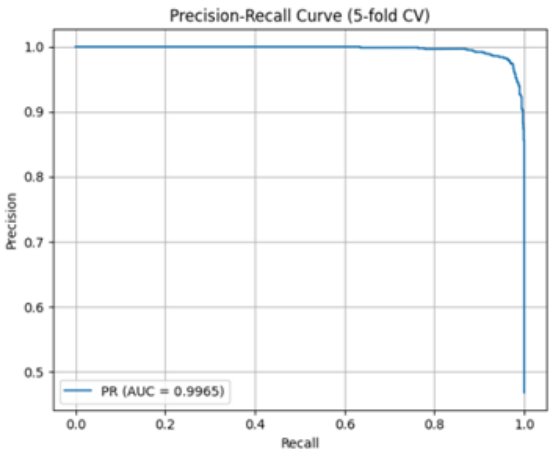


Figure 3.3: ResNet50

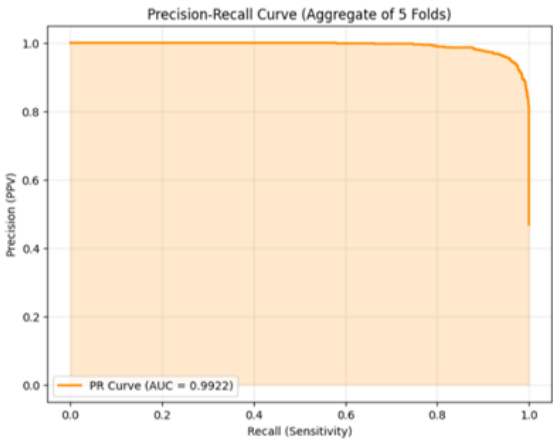


Figure 3.4: EfficientNetB0

3.4 Qualitative Observations

The custom CNN required more epochs to converge and showed greater validation loss fluctuations. DenseNet121 and ResNet50 converged faster, whereas EfficientNetB0 provided the best balance between speed and accuracy.

4. Result Discussion

The experimental results show that transfer learning models consistently outperform the custom CNN in most evaluation metrics for chest X-ray based viral pneumonia classification. Among all models, ResNet50 achieved the most balanced performance with an accuracy of 97.3%, stable precision-recall values, and a ROC-AUC of 0.992, indicating strong discriminative capability.

DenseNet121 also demonstrated competitive performance with 96.2% accuracy and high recall (97.5%), reflecting its strong feature reuse mechanism through dense connections. However, its accuracy remains slightly lower than ResNet50, possibly due to sensitivity to dataset-specific variations.

EfficientNetB0 achieved the highest ROC-AUC of 0.994 and excellent precision and recall values (98.0% and 98.1% respectively), showing strong classification confidence. However, its reported accuracy (95.3%) is comparatively lower than other transfer learning models, which may indicate variability in decision threshold sensitivity or fold-wise performance fluctuations.

The custom CNN performed the weakest overall, achieving 94.3% accuracy and lower recall compared to other models. This confirms that training from scratch is less effective for medical imaging tasks where dataset size is limited and feature complexity is high.

Overall, the results confirm that pre-trained deep learning models provide more robust and reliable feature extraction for pneumonia detection, with ResNet50 and EfficientNetB0 emerging as the most effective architectures in this study.

4.1 Error Analysis

Most false positives occurred in images with mild opacities or atelectasis that resembled pneumonia. False negatives mostly occurred in early-stage pneumonia cases with subtle interstitial markings.

4.2 Strengths and Limitations

4.2.1 Strengths

- Rigorous 5-fold cross-validation
- Inclusion of clinically meaningful metrics such as NPV
- Fair comparison under identical training conditions
- Use of modern training techniques

4.2.2 Limitations

- Dataset obtained from a single source
- No external validation
- Binary classification only
- Lack of explainability methods such as Grad-CAM

4.3 Future Work

Future directions include:

- Multi-class classification
- External validation on larger datasets
- Explainability using Grad-CAM
- TensorFlow Lite deployment
- Ensemble learning approaches

5. Conclusion

This study presented a comparative evaluation of a custom CNN and three transfer learning models for viral pneumonia detection from chest X-ray images using 5-fold cross-validation.

The results demonstrate that transfer learning significantly improves performance over training a model from scratch. ResNet50 achieved the most balanced overall performance, and showed the highest ROC-AUC and strong classification confidence.

The custom CNN showed comparatively lower performance, confirming the limitations of non-pretrained architectures in medical image analysis.

Overall, transfer learning models, particularly ResNet50 and DenseNet121, provide more reliable and accurate solutions for automated pneumonia detection systems.

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